



# ANNOUNCING PARAMETER

13

$$\frac{\mathrm{d}}{\mathrm{d}\beta} S(\beta) = k_B \langle U \rangle_\beta + k_B \frac{\mathrm{d}}{\mathrm{d}\beta} \langle U \rangle_\beta + k_B \frac{\mathrm{d}}{\mathrm{d}\beta} \log Z(\beta)$$

► **Proposition:** It denotes the increasing in entropy for an increase in energy average energy

$$\frac{dS}{d\langle U \rangle} = k_B \beta = \frac{1}{T}$$

► **Proof:** Recall that the derivative of the energy equals

► If we take the derivative of the energy w.r.t.  $\theta$ , we get:

$$\langle U \rangle_\beta = - \frac{\mathrm{d}}{\mathrm{d}\beta} \log Z(\beta), \quad \frac{\mathrm{d}}{\mathrm{d}\beta} \langle U \rangle_\beta = - \frac{\mathrm{d}^2}{\mathrm{d}\beta^2} \log Z(\beta) := I(\beta)$$

==

—

*KBIB*

► If we take the derivative of the energy equation:

$$\frac{\mathrm{d}S}{\mathrm{d}\langle U \rangle} = \frac{\mathrm{d}S}{\mathrm{d}\beta} \bigg/ \frac{\mathrm{d}\langle U \rangle}{\mathrm{d}\beta} = k_B \beta$$

- **Proposition:** It denotes the increasing in entropy for an increase in energy average energy

$$\frac{dS}{d\langle U \rangle} = k_B \beta = \frac{1}{T}$$

- **Proof:** Recall that the derivative of the energy equals satisfies:

$$\langle U \rangle_\beta = -\frac{d}{d\beta} \log Z(\beta), \quad \frac{d}{d\beta} \langle U \rangle_\beta = -\frac{d^2}{d\beta^2} \log Z(\beta) := I(\beta)$$

- If we take the call that the derivative of the energy equals satisfies:

$$\frac{d}{d\beta} S(\beta) = k_B \langle U \rangle_\beta + k_B \frac{d}{d\beta} \langle U \rangle_\beta + k_B \frac{d}{d\beta} \log Z(\beta) = -k_B I(\beta)$$

- If we take the call that the derivative of the energy equals satisfies:

$$\frac{dS}{d\langle U \rangle} = \frac{dS}{d\beta} \bigg/ \frac{d\langle U \rangle}{d\beta} = k_B \beta$$

# TEMPERATURE